



Zooplankton response to eutrophication in multiple aquatic habitats using remote sensing approach

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Abstract

Ten surface water samples were collected from five distinct aquatic environments; the Mediterranean Sea, Estuary, Barge Canal, Nile River and Manzala Lake. Each water body possesses unique ecological characteristics. Zooplankton communities were analyzed across these environments to assess biodiversity and eutrophication status. Results revealed that trophic conditions varied significantly according to the type of water body. Mesotrophic conditions dominated the central part of the study area around the Nile River, while eutrophic conditions prevailed in the southeastern region near Manzala Lake. A total of 23 zooplankton species were recorded and classified into seven major groups. The Mediterranean Sea exhibited the highest biodiversity (11 species), whereas the Barge Canal had the lowest (4 species). Manzala Lake showed the highest standing crop (29520 org/m³), reflecting its nutrient-rich and highly productive environment influenced by agricultural and industrial discharges. This elevated nutrient input promotes marine enrichment and enhances zooplankton diversity, particularly of brackish and euryhaline species. Crustaceans dominated the Mediterranean Sea, highlighting their importance in the marine food web, while meroplanktonic larvae were predominant in the Barge Canal, indicating its function as a spawning zone for benthic organisms. Crustaceans and meroplankton were also dominant in the Nile Delta coastal area and Manzala Lake, reflecting favorable conditions for brackish water communities. Rotifers were the most prevalent group in the Nile River, indicating potential eutrophication and anthropogenic pressure.

Keywords Damietta governorate · Phytoplankton blooming · Manzala lake · River Nile

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Extended author information available on the last page of the article

Introduction

Water resources policy at all levels needs to be regularly assessed due to the serious global issue of water pollution, especially in Egypt. (Okbah et al. 2024; El-Kowrany et al. 2016). Due to export of residual wastes to the coastal areas which are anthropogenic activities, they can include: agriculture (fertilizers), aquaculture (fish feeds), industrialization (salts), social-cultural norms (organic wastes) and atmospheric inputs (Iqbal et al. 2023; Song et al. 2016). The elevated nutrient levels in the water body are caused by the accumulation of residual wastes. The excessive concentrations of chlorophyll-a (Chl-a), total phosphorus (TP) and total nitrogen (TN) indicate an excess of nutrients and lead to eutrophication, which in turn creates circumstances of oxygen deprivation and hazardous algal blooms, ultimately undermining the water body's ability to carry water (Saputra et al. 2017). The eutrophication process is a natural one in increasing production of organic materials accompanying the evolution of an aquatic ecosystem over geologic time, until eventually it fills up completely (Le Moal et al. 2019). Eutrophication is a widespread problem in rivers, lakes, estuaries and coastal area (El-Amier et al. 2016). Recent studies have provided a comprehensive evaluation of the trophic status of Lake Manzala, confirming the continuous impact of anthropogenic discharges and the need for sustainable management (Hegab et al. 2024). Manzala Lake and its surrounding areas, including the river, Barge Canal and estuaries are of great importance as a Mediterranean lagoon in Egypt, receiving substantial pollutant loads from multiple drainage sources. These pollutants give rise to serious issues for Lake Manzala, such as a reduction increased eutrophication and decreased water quality; in addition to the fish output has decreased. So, starting in 2017, the Egyptian government performed a lot of efforts to restore natural ecosystem of this important lake (El-Mezayen and Abd El-Hamid 2023). Zooplankton plays a vital role transferring energy in aquatic ecosystems between primary producers and planktivorous organisms (Santos-Wisniewski et al. 2006). They are considered bioindicators due to their sensitivity to various factors, such as surface area, water current, depth and pollution. Sarnelle (2007) suggests that *Daphnia* species can effectively manage phytoplankton bloom, resulting in an increase in other smaller species of cladocerans. Eutrophication in water bodies results from excessive blooming of phytoplankton communities. This blooming can be toxic and disrupts the food web, leading to a shortage of food for grazers (Paerl and Paul 2012). The rapid life cycle of zooplankton suggests that increased blooms driven by eutrophication may enhance physiologically tolerant species by promoting their growth as defense mechanism against eutrophication (Ger et al. 2014). Sarnelle (2007) suggests that *Daphnia* species can effectively manage phytoplankton bloom, resulting in an increase in other smaller species of cladocerans. The relationship between zooplankton and phytoplankton biomass is very complicated and may be due to data limitations; data sometimes collected from eutrophic environments, or sometimes from oligotrophic environments, or from multiple studies (Jeppesen et al. 2003; Hessen et al. 2006; Heathcote et al. 2016). Additionally, climate change or humane activity may also play a vital role (Keizi 2006; Selas 2003). Remote sensing and GIS are a precisely accurate and low-cost technique (Abd El-Hamid 2020). Remote sensing (RS) techniques have become powerful tools for monitoring and assessing water quality parameters in both inland and coastal environments. Satellite-based sensors such as Landsat OLI, Sentinel-2 MSI, and MODIS provide multispectral and hyperspectral data that allow the estimation of water constituents, including chlorophyll-a and cyanobacterial blooms (Liang et al. 2017).

These parameters are closely linked to eutrophication and trophic state variations (Kutser et al. 2016; Johansen et al. 2018). Remote sensing also supports the identification of pollution sources, algal bloom dynamics, and nutrient loading patterns, enabling cost-effective large-scale monitoring of aquatic systems (Laneve et al. 2024). The application of RS in trophic state modeling offers significant advantages in terms of coverage, frequency, and accessibility, especially in areas where in situ sampling is logistically difficult (El-Mezayen and Abd El-Hamid 2023). Therefore, coupling satellite observations with ground-based data provides an efficient approach to evaluate water quality and detect early signs of eutrophication. The present study aims to assess the trophic status and zooplankton community diversity across multiple aquatic habitats with distinct environmental characteristics. Emphasis was placed on integrating in-situ environmental measurements with remotely sensed data to develop a comprehensive assessment model. This integration enhances the understanding of how eutrophication levels and environmental variations influence zooplankton distribution and community structure. Remote sensing data were employed as a complementary tool to laboratory analyses, allowing spatial evaluation of eutrophication patterns and supporting the identification of ecological responses across different water bodies.

Materials and methods

Study area

This study area includes different water bodies; Sea, Estuary, Barge canal, Nile River and Manzala lake. Each water body has a different property. Sea water (Mediterranean Sea), brackish water (Manzala Lake), Barge canal (Mixed between fresh and marine), Estuary (moderate salinity) and fresh water (Nile River). The trophic state of the study area is primarily influenced by human activities such as industrial production, the use of fertilizers and pesticides and urban sewage discharge. These activities lead to excessive accumulation of nutrients in the soil and water, which then flow into the lake and gradually worsen eutrophication. The study area extends from New Damietta to Manzala Lake. The predominant wind direction at northeastern coast of Nile Delta is between N and NW direction with an average wind speed of about 3.75 m s^{-1} (Khalifa et al. 2017). Sampling sites were selected to represent the various water bodies within the study area, as illustrated in Fig. 1.

Field measurements

Water samples were collected covering three different environments. Ten surface water samples were collected during November 2021. The sample time span period corresponds to the acquisition times of Landsat (OLI) remote sensing image. Samples were put into polyethylene bottles, returned to the laboratory and stored inside a dark refrigerator at 4°C for further analyses. Coordinates of each sampling site were recorded using a GPS device and presented in degrees and minutes format ($^\circ$ ') as shown in Table 1. Surface water temperature and salinity were taken in the field. For determination of dissolved oxygen (DO) mg/l, samples were fixed in the field and Winkler's method was adopted as described in APHA (2005), including ammonia (NH_4) determination. Samples were transferred to the laboratory; samples were filtered using $0.45 \mu\text{m}$ membrane filter papers for determination

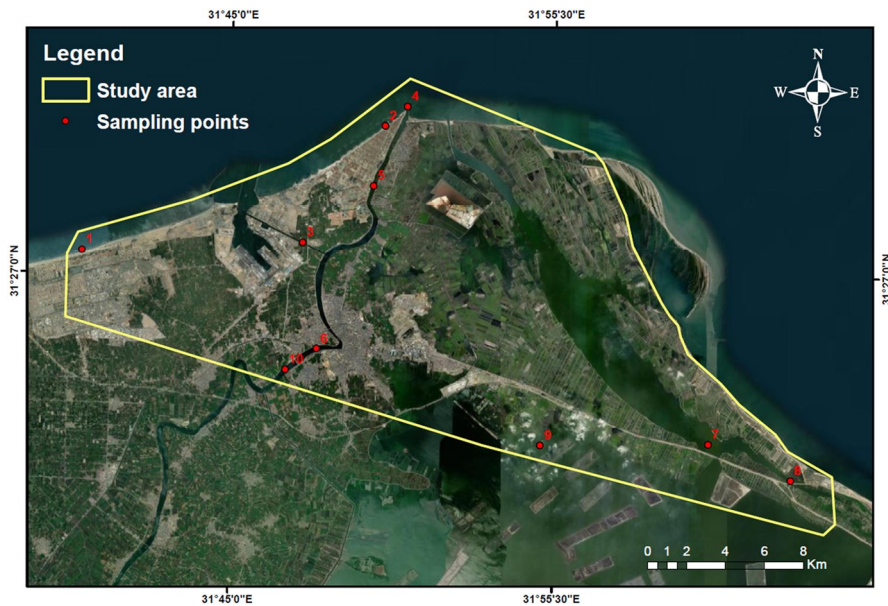


Fig. 1 Sampling locations from studied different water bodies

Table 1 Sampling locations (Latitudes and longitudes) for studied water bodies

S. No.	Location	N (Latitude)	E (Longitude)
1	Sea 1	31° 40'	31° 28'
2	Sea 2	31° 50'	31° 31'
3	Barge Canal	31° 47'	31° 28'
4	Estuary 1	31° 51'	31° 32'
5	Estuary 2	31° 49'	31° 29'
6	Estuary 3	31° 48'	31° 25'
7	Manzala Lake 1	32° 00'	31° 22'
8	Manzala Lake 2	32° 03'	31° 21'
9	Manzala Lake 3	31° 55'	31° 22'
10	Nile River	31° 47'	31° 24'

of some nutrients as NO_2 , PO_4 , NO_3 , total nitrogen, total phosphorus and SiO_4 according to Grasshoff et al. (1999). For determining the phytoplankton biomass as chlorophyll (a) and chlorophyll (b) a known volume of the water samples was filtered using nitrocellulose membrane filter papers of 0.45- μm pore size. Then filtered paper containing chlorophyll was extracted into 90% acetone and then measured at a spectrophotometer and finally calculated according to Strickland and Parsons (1972) using the score UNESCO equation: $[(11.64 * E_{663}) - (2.16 * E_{645}) + (0.1 * E_{630})] * 10/V$. The final results were expressed as microgram per liter.

Remote sensing data

Landsat imagery with minimal cloud cover was acquired from the U.S. Geological Survey website (<https://glovis.usgs.gov>) in November 2021. QGIS 16.3 software was used for pre processing and image enhancement. The conversion of digital number (DN) values to top-of-atmosphere reflectance (TOA) and subsequent atmospheric correction were performed using the dark object subtraction (DOS) algorithm (Chavez 1988). The feasibility of the trophic state index (TSI) inversion algorithm varies among different sensors due to their spectral configurations (Johansen et al. 2018). However, Sentinel-2/MSI provides three special spectral bands (vegetation red edge) that are particularly suitable for monitoring trophic state changes (Kutser et al. 2016). Similar approaches have been successfully applied to monitor phytoplankton dynamics and bloom development in Mediterranean coastal systems (Soriano-González et al. 2019). Ordinary kriging interpolation was applied in ArcGIS 10.8 to generate continuous spatial maps of environmental parameters from the limited number of sampling points (Abd El-Hamid et al. 2023). These remotely sensed parameters (e.g., surface temperature, chlorophyll-a concentration, and suspended matter derived from Landsat data) were further correlated with in-situ zooplankton density measurements to assess the relationship between water quality variations and zooplankton distribution in the investigated water bodies.

Eutrophication status

The Trophic State Index (TSI) was estimated using a combination of measured environmental parameters including total nitrogen (TN), total phosphorus (TP), turbidity, and chlorophyll-a (Chl-a). These parameters were integrated following the empirical models of Carlson (1977) and subsequent modifications by Jeppesen et al. (2003) and Le Moal et al. (2019). For each parameter, individual TSI values were calculated and averaged to obtain a composite index representing the trophic status of each site. Furthermore, a multiple regression model was applied to predict TSI based on water-quality variables (NO_2 , PO_4 , SiO_4 , TN, and TP) following the approach of El-Mezayen and Abd El-Hamid (2023) and Hegab et al. (2024). The resulting TSI values were classified into oligotrophic, mesotrophic, eutrophic, and hypertrophic categories according to Carlson's (1977) thresholds. As calculated in next equations:

$$\text{TSI}(\text{ch}_a) = 9.81 * \ln \text{ch}_a + 30.6 \quad (1)$$

$$\text{TSI}(\text{TP}) = 14.42 * \ln \text{TP} + 4.15 \quad (2)$$

$$\text{TSP}(\text{TN}) = 54.45 * \ln \text{TN} - 19.24 \quad (3)$$

$$\text{TSP}(\text{Turbidity}) = 25.2 * \ln \text{Turbidity} - 10.5 \quad (4)$$

$$\text{TSI} = \frac{\text{TSI}(\text{ch}_a) + \text{TSI}(\text{TP}) + \text{TSP}(\text{TN}) + \text{TSP}(\text{Turbidity})}{4} \quad (5)$$

where: Chl-a is the chlorophyll-a concentration ($\mu\text{g L}^{-1}$), TP is the total phosphorus concentration ($\mu\text{g L}^{-1}$), TN is the total nitrogen concentration ($\mu\text{g L}^{-1}$), and Turbidity is measured in nephelometric turbidity units (NTU). The final TSI value was calculated as the average of the individual indices, representing the overall trophic condition of the investigated water bodies.

Prediction of eutrophication using TSI

Linear regression model was applied in the present study to predict eutrophication status based on TSI. For accurate prediction of eutrophication, correlation was applied among all parameters both measured and remote sensing data. The accuracy of the predicted model was examined based on R^2 .

Zooplankton sampling and preservation

Zooplanktons were collected using plankton net with 45 Cm mouth diameter and 45- μm mesh size, performing a slow vertical haul from the bottom to the surface of the water column. The filtrated samples were concentrated and transferred into polyethylene bottle. To preserve sample, 10% neutral formalin was added immediately. The samples were left in place for an adequate period to ensure complete settling of all organisms. The zooplankton samples then concentrated to 100 ml volume. The standing crop of total zooplankton was estimated by taken a 5 ml subsample after gently shaking the container and transferred it to a Bogorove chamber using a plunger pipette. The number of zooplankton was calculated as a number of organisms per litre (organisms/ m^3) using the standard formula for concentrated samples (APHA, 2017):

$$N = (n \times V_t) / (V_s \times V_f).$$

Where:

N = number of Organisms per cubic meter (organisms/ m^3).

n = number of individuals counted in the sub-sample.

V_t = total volume of concentrated sample (ml).

V_s = volume of the sub-sample analyzed (mL).

V_f = volume of water filtrated by zooplankton net (m^3).

To prevent degradation and maintain organismal integrity. In the laboratory, the samples were gently homogenized and sub-sampled using a Stempel pipette for quantitative and qualitative analysis. All zooplankton organisms were identified and counted at the species level in three 5 ml replicate. The average count from these replicates was expressed as organisms/L. for identification, various publications and taxonomic references were used, including Rose (1933); Tregouboff and Rose (1957); Edmondson (1959); Marshall (1969); Bick (1972); Cosper (1972); Harding and Smith (1974); Balcer et al. (1984); Shiel and Koste (1992); Witty (2004).

Table 2 Average values for water quality parameters along different studied water bodies

St. No	pH	Salinity ‰	Turbidity	NTU	DO (mg/L)	Ammonia (µg/L)	Nitrite (µg/L)	Nitrate (µg/L)	Phosphate (µg/L)	Silicate (µg/L)	Chl. a (µg/L)	TN (µg/L)	TP (µg/L)
1	8.27	38.7	51		8	127.92	27.21	66.21	7.04	0.553	133.93	611.8	80.85
2	8.22	41.5	12.6		9.4	27.06	18.46	34.41	16.23	0.559	88.17	511.98	38.85
3	7.82	6.7	3.8		9.3	104.55	67.57	103.34	8.27	0.662	8.81	663.32	119.7
4	7.91	19.2	21		8.7	57.81	43.79	58.32	19.3	0.902	41.3	1268.68	72.45
5	7.87	7.9	5.4		6.5	110.7	76.33	104.72	6.43	0.673	44.19	924.14	135.45
6	7.69	4.5	6.8		9.3	52.89	66	150.54	4.29	0.808	36.56	888.72	139.65
7	7.94	12.6	37		8.2	35.67	23.46	58.38	6.13	2.157	270.05	1626.1	207.9
8	8.55	19.2	33		7.7	82.41	35.66	101.22	6.43	1.907	128.24	1693.72	136.5
9	8.66	12	36		7.8	51.66	15.95	86.89	30.02	1.506	225.39	1590.68	309.75
10	7.27	0.5	6.1		9.9	33.21	71.32	282.09	49.31	0.716	141.3	888.72	87.15

Results and discussion

Water quality

Almost each biological population in the water body is impacted by eutrophication, which has a negative impact on the biological stability of the water body itself (Ahmed et al. 2006). From Table 2, pH values along the five studied water bodies ranged within the alkaline zone, recording a minimum of 7.27 at the Nile River and a maximum of 8.66 at Manzala Lake 3. These variations in pH may result from both natural processes, such as alkaline geology, and anthropogenic influences including industrial effluents, agricultural runoff, and urban wastewater (Vasistha and Ganguly 2020; Okbah et al. 2024). As shown in Table 2, the selected water bodies can be classified according to their salinity into distinct categories: the Nile River represents oligohaline water (0.5‰), Estuary 3 (4.5‰), and the Barge Canal (6.7‰) fall under mesohaline conditions. Higher salinity values were observed at Manzala Lake 1 and 3 (12.7‰ and 12‰, respectively) and at Estuary 1 and Manzala Lake 2 (19.2‰), which correspond to polyhaline conditions. The Mediterranean Sea sites (Sea 1 and Sea 2) showed euhaline salinity levels (38.7‰ and 41.5‰, respectively). These findings reflect the natural hydrological characteristics of each water body, where Manzala Lake represents brackish water, the Barge Canal exhibits mixed fresh and marine influence, and the Estuary shows intermediate salinity (Lembi 2001; Kültz 2015). The variation in pH, salinity, and nutrient levels reflects the influence of both natural and anthropogenic factors, consistent with findings from Lake Pamvotis, Greece (Kagalou et al. 2001). Table 3 shows the lowest value of turbidity was found at location of Barge canal (3.6 NTU) but the highest one was at Sea 1 location (51 NTU) passing through the intermediate one of Estuary 1 location (21 NTU), turbidity reflects the health of any water body via transparency of this body. In most cases of water bodies, high turbidity values may be a sign for eutrophication situation (Yang et al. 2008). DO is the concentration of oxygen gas in the water which is absorbed directly from air or from photosynthesis processes by aquatic plants. Decreasing in DO value may cause a death of aquatic fish (Müller et al. 2012). It is noticed from Table 3 that DO values ranged from 6.5 mg/l Estuary 2 location to 9.9 mg/l at Nile River location. Results showed that all values are within the permissible levels provided by WHO (2017). This indicates that these values of DO are adequate for most aquatic life (Gosomji and Okooboh 2013). Nile river location recorded for the highest value of nitrate (282 µg/l) while location of Sea 2 recorded the lowest value of (34.41 µg/l). Ammonia results recorded the highest value at location of Sea 1 (127 µg/l) which means for recent organic pollution, but Sea 2 location recorded the lowest value (27.06). High values of pollution were observed in coastal marine water and the nearby estuary (El-Alfy et al. 2019). Nitrite as an intermediate state between nitrate and ammonia highest value was recorded at estuary 1 (76.33 µg/l) but the lowest value was at Manzala lake 3 location (15.95 µg/l). Nile river location recorded for the highest value of nitrate (282 µg/l) which means an old organic pollution while location of Sea 2 recorded the lowest value of (34.41 µg/l). Phosphate highest value (49.31 µg/l) was recorded at the same one of nitrate location of Nile River, may be resulted from usage of fertilizers in agricultural areas surrounded this location and enrich water bodies with run-off from agricultural farms, this could help in progress of eutrophication process (Yang 2022). The highest silicate value was recorded at the Manzala Lake 1 location (2.157 µg/l); this may be due to discharge of agricultural wastes. On the other hand, the lowest value of silicate was at Sea 1 loca-

Table 3 Status of eutrophication based on previous parameters

TSI Degree	State	Description
<40	Oligotrophic	Low productivity, clear water and low algal biomass
40–50	Mesotrophic	Moderate productivity, intermediate levels of nutrients and algal biomass
50–70	Eutrophic	High productivity, high levels of nutrients and high algal blooms
>70	Hypereutrophic	Very high productivity, severe algal blooms and low oxygen

tion (0.553 $\mu\text{g/l}$). For chl *a*, the highest value was recorded at the Manzala Lake location (270.05 $\mu\text{g/l}$) and the lowest value was at Barge canal location (8.81 $\mu\text{g/l}$). This indicator for enhancing of eutrophication process at these locations. Figure 2 displays the image-derived maps for surface water turbidity, pH, salinity, dissolved oxygen, ammonia, nitrite, nitrate, phosphate, silicate, total nitrogen, total phosphorus and chlorophyll *a*. Data of pH show that the higher values around Manzala lake and sea but the lower values near to Nile River. Higher pH may be as a result of biological activity and minerals in these zones. Salinity data show high differences among water bodies as salinity increase near to sea reaches to 41‰ but it reaches to 0.5‰ in Nile River. For DO, the red color indicates regions with higher dissolved oxygen are beneficial for aquatic organisms' survival, where the blue color relates to zones of low oxygen that cause stress on aquatic life. For ammonia, the red color reveals that these areas are sensitive to pollution as decomposition of organic matter; on the other hand blue color indicates that these areas have a better water quality or less pollution. For silicates, the red color indicates to high concentrations of silicates as a result of runoff or agriculture wastes near to Manzala lake where the low concentrations around to sea and Nile River.

Eutrophication status

In the present study, the Trophic State Index (TSI) was calculated using turbidity, total nitrogen (TN), total phosphorus (TP), and chlorophyll-*a* (Chl-*a*) data. These parameters were used to classify the eutrophication status of the investigated water bodies into four trophic levels: oligotrophic, mesotrophic, eutrophic, and hypertrophic, as summarized in Table 3. From last findings, situation of eutrophication process at the selected water bodies at this study could be summarized as: Both Nile River and Manzala Lake locations have moderate situation but Barge canal and Sea 2 location has slightly situation. Sea 2 location has highly situation of eutrophication process. Depending on TSI model, the measured TSI show that distribution of oligotrophic covers a significant portion of the northwestern near to Nile River and Barge canal along the study area where, the mesotrophic cover the center and final the eutrophic covers the southeastern part around Manzala Lake as shown in Fig. 3. Various mitigation strategies have been proposed to control eutrophication, including nutrient load reduction and ecological restoration (Haijun and Hongzhu 2009). On the other hand, the predicted image of TSI shows that the oligotrophic cover a large area compared to the measured TSI, where the mesotrophic confined to the south compared to the measured TSI. Similar modeling approaches for evaluating eutrophication and water quality have been successfully applied in other Mediterranean lakes, such as Lake Pamvotis (Yannopoulos

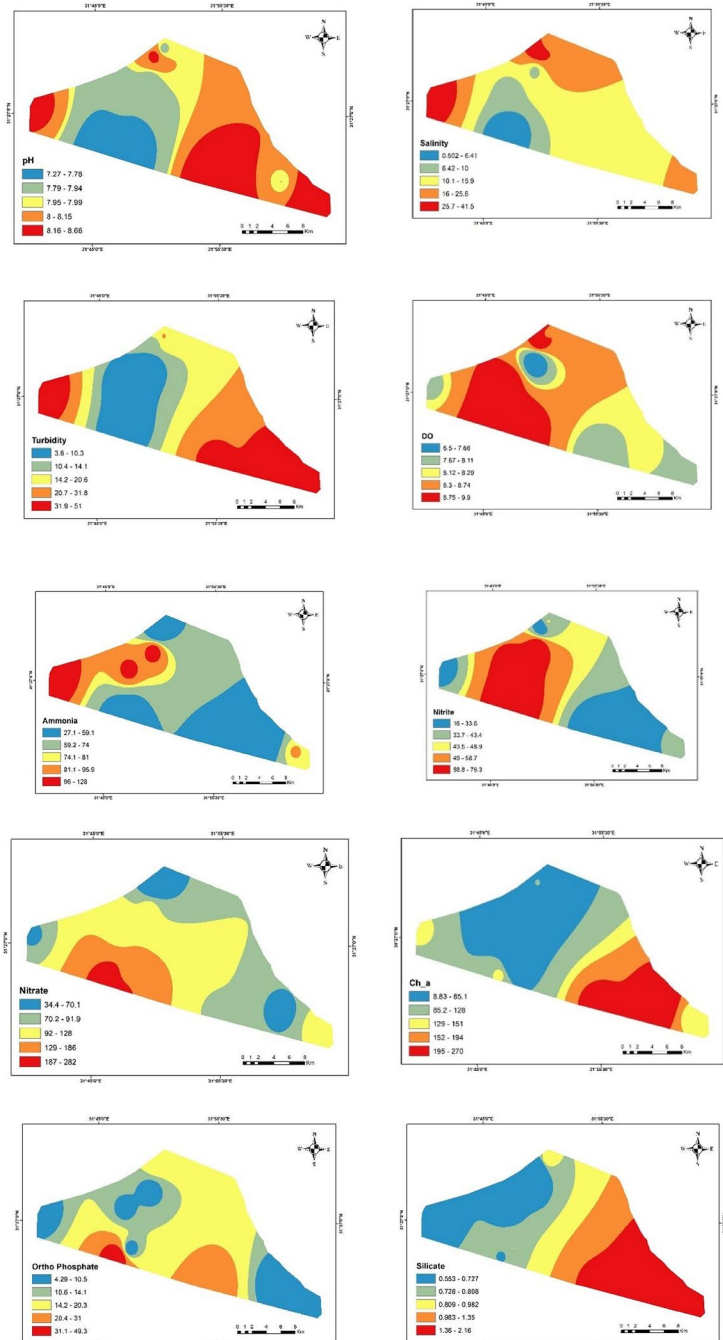


Fig. 2 Spatial distribution of water quality parameters along the study area

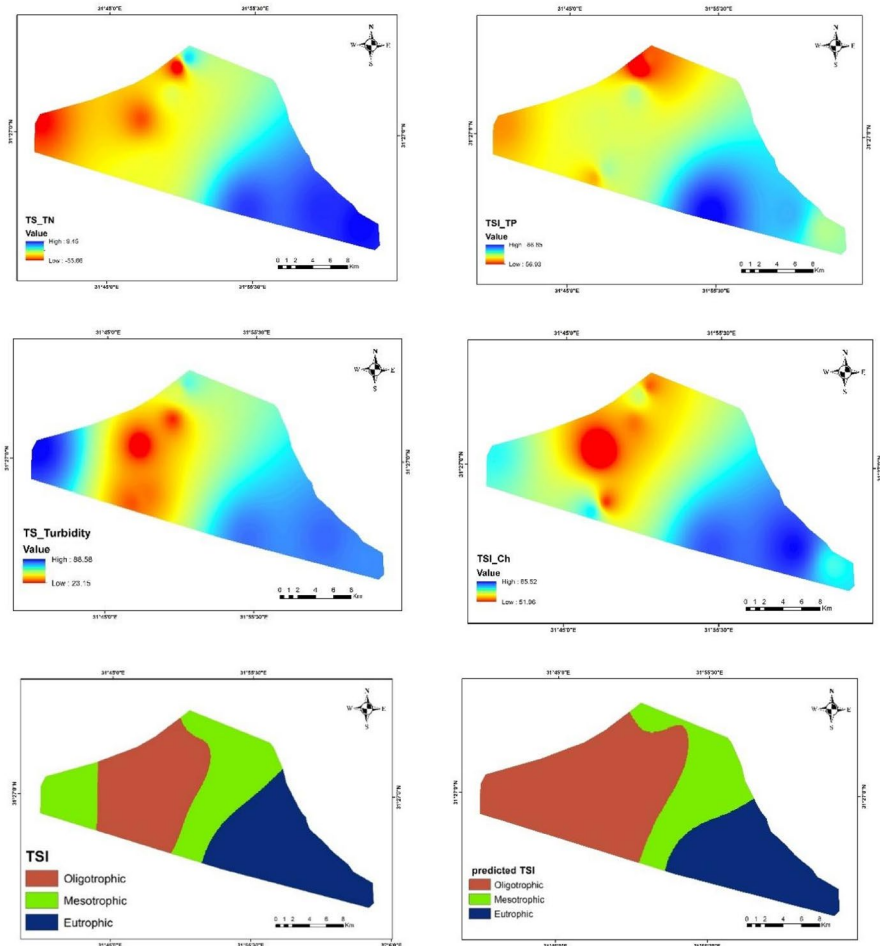


Fig. 3 Spatial distribution of Eutrophication indices along the study area

and Kaloyannis 2008). Finally, the eutrophic of predicted TSI appears to be smaller than the measured TSI. The predicted TSI was extracted using the equation with R^2 equal to 0.78. The smaller portion of eutrophic may be related to recent development in Manzala Lake.

$$\text{TSI} = 29.629 - 0.2 * \text{Nitrite} + 0.011 * \text{Phosphate} - 1.814 * \text{Silicate} + 23.40 * \text{TN} + 0.0136 * \text{TP}$$

This empirical equation was developed to predict the trophic state index (TSI) from nutrient concentrations. Nitrite (NO_2), phosphate (PO_4), silicate (SiO_4), total nitrogen (TN), and total phosphorus (TP) represent key eutrophication drivers. Positive coefficients (TN and TP) indicate increasing eutrophication, while negative ones (NO_2 and SiO_4) show inverse effects. The model followed the regression approach used by El-Mezayen and Abd El-Hamid (2023).

Table 4 Zooplankton list collected from five habitats of Damietta Governorate, Egypt

Protozoa	Copepoda
Amoebozoa	<i>Acartiaclausi</i> (Giesbrecht, 1889)
<i>Centropexisecornis</i> (Ehrenberg, 1841)	<i>Oithona nana</i> (Giesbrecht, 1893)
Tintinnida	<i>Euterpinaacutifrons</i> (Dana, 1847)
<i>Favella serrata</i> (Möbius, 1887)	Copepodite stage
<i>Leprotintinnusnordqvistii</i> (Brandt, 1906)	Nauplius larvae
<i>Tintinnopsisberoidea</i> (Stein, 1867)	Ostracoda
<i>Tintinnopsis cylindrica</i> (Daday, 1887)	<i>Candonasubgibba</i> (G. O. Sars, 1926)
Foraminifera	<i>Cypridinamediterranea</i> (O. Costa, 1845)
<i>Ammonia</i> sp	Amphipoda
Nematoda	<i>Gammaruslacustris</i> (G.O. Sars, 1864)
<i>Rhabdolaimus minor</i> (Cobb, 1914)	Mycetaceae
Rotifera	<i>Mysis relicta</i> (Lovén, 1862)
<i>Asplanchniapriodonta</i> (Gosse, 1850)	Chaetognatha
<i>Brachionusangularis</i> (Gosse, 1851)	<i>Sagittafrederici</i> (Ritter-Záhony, 1911)
<i>Brachionusplicatilis</i> (Müller, 1786)	Appendicularia
<i>Filinalongiseta</i> (Ehrenberg, 1834)	<i>Oikopleuradioica</i> (dioica Fol, 1872)
<i>Keratellavalga</i> (Ehrenberg, 1834)	Meroplanktonic larvae
<i>Polyarthra vulgaris</i> (Carlin, 1943)	Cirriped larvae
<i>Synchaetaoblonga</i> (Ehrenberg, 1832)	Trochophore larvae

Community structure of zooplanktons

A total of 23 species (Table 4) were recorded and identified to the nearest genera with maximum biodiversity of 11 species at the Mediterranean Sea and minimum value of four species at the Barge Canal. The maximum standing crop value was recorded at Manzala Lake (295.2 organisms/L) and the minimum was at the Barge Canal (15.6 organisms/L) (Fig. 4). As shown in Fig. 5, zooplankton composition differs between the five studied habitats, reflecting the salinity, and nutrients loads. In the Mediterranean Sea coast, crustacean was the dominant (69.44%), that may be due to copepods which are the most prevalent organisms for the food web and serve as the main fish food in marine habitat (Feng et al.2023). The second most abundant group was Protozoa which are highly effective bioindicators for water pollution in marine habitat (Mostafa et al.2023). Rotifera, appendicularians, chaetognaths and meroplanktonic larvae were recorded, but with minor contribution, indicate the presence of both predatory and temporary larvae of benthos. In the Barge Canal meroplanktonic larvae is the dominant, makes up (65.38%) of the total zooplankton community, suggested that the Barge Canal is a suitable habitat as spawning zone for benthos, indicating the influence of anthropogenic factors from shipping activities which transfer species from different habitats (Ruiz et al. 1997). Also, crustaceans which were an important grazer in marine habitat represented by (26.29%) of the total zooplankton of the Barge Canal (Heck

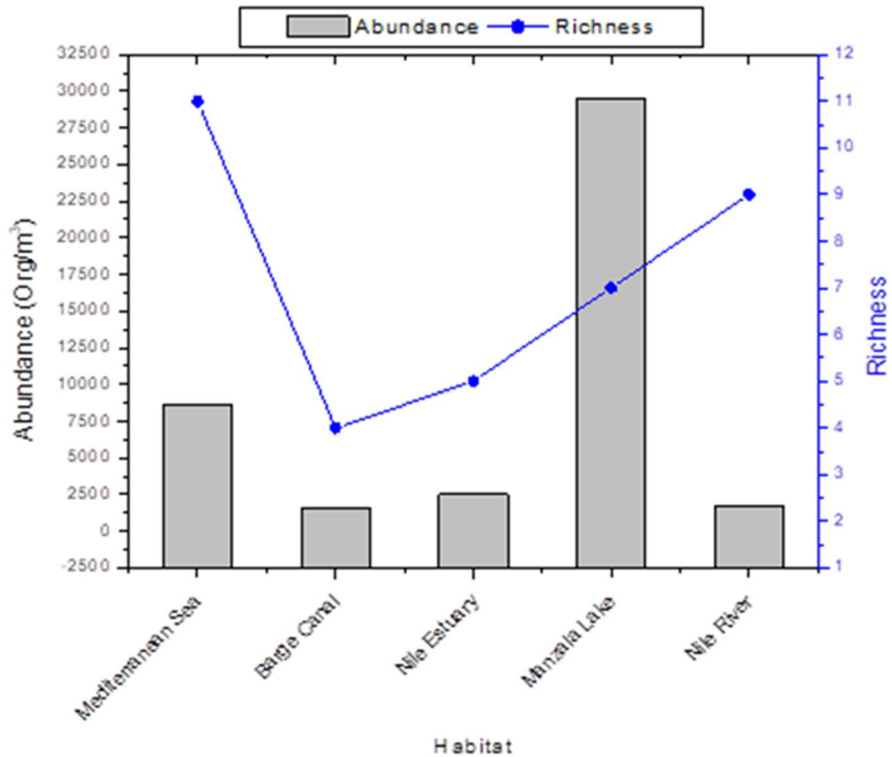


Fig. 4 Abundance and richness of zooplankton in five habitats along Damietta Governorate

and Valentine 2007). Rotifers represented by (7.69%), indicating some microbial and/or eutrophic conditions. Crustaceans (45.24%) and meroplanktons (40.48%) dominate, with notable share of rotifers (7.14%) and protozoans in the Nile Delta coastal area, reflecting the transitional nature of the Nile Delta coastal area. Its nutrient rich habitat supporting both marine and freshwater species. Meroplanktons existence explains that Nile Delta coastal area serve as nursery area for brackish water species in a part of their development (Beck et al. 2001). Rotifers and protozoans indicated also some microbial and/or eutrophic proliferation. Crustaceans (46.75%) and meroplanktons (40.65) again were dominated in Manzala Lake, with contribution of protozoans (9.35%), rotifers (3.05%) and nematodes (0.2%). The zooplankton structure of Manzala Lake reflect their nature as semi-closed system mixed both fresh and saline water which provide a fertile environment for grazers and meroplanktons. Also, the existence of protozoans, rotifers and nematodes ensure environmental shifts like eutrophication and microbial blooming which may resulting from anthropogenic and other human harmful activities (Pratiwi et al. 2016). Zooplankton structure of Nile River of four groups; rotifers (42.86%), crustaceans (21.43%), meroplanktons (32.14%) and protozoans (3.57%). The high rotifers dominance and less abundance of crustaceans reflects the typical freshwater environments for rivers which suitable for rotifers and less suitable for crustaceans (Segers 2007).

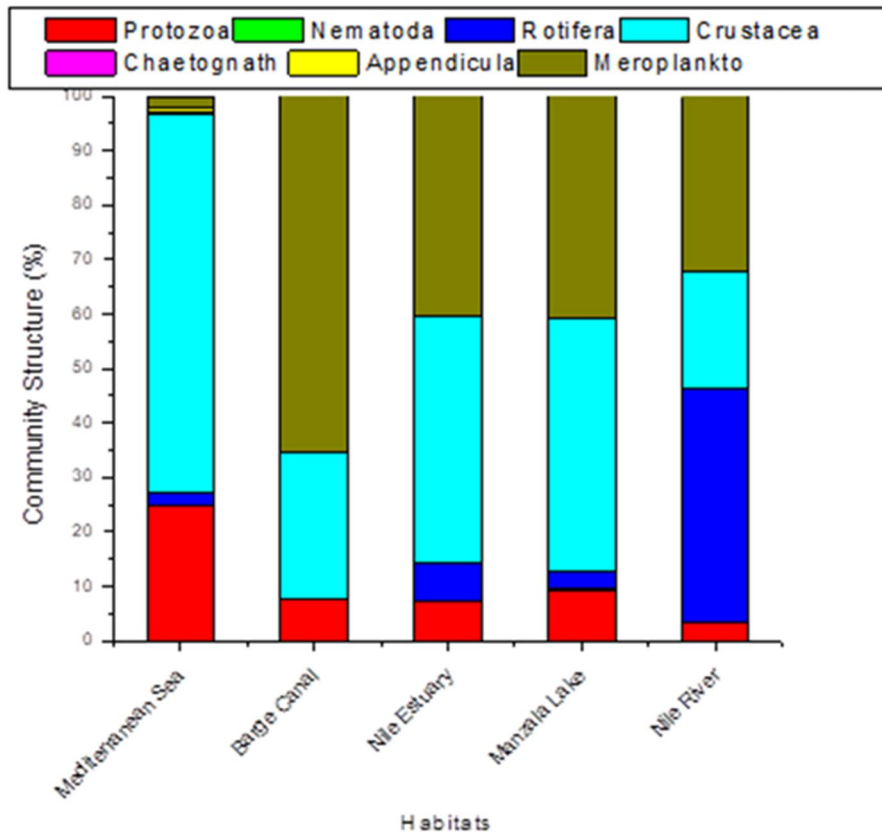


Fig. 5 Zooplankton community structure in five different habitats of Damietta Governorate

Statistical analysis

Correlation of all parameters and species

There is a strong positive correlation between Ammonia and Nitrite with a value of 0.77, as well as with Phosphate with a value of 0.63. Turbidity is positively correlated with Dissolved Oxygen (DO) with a value of 0.77, while it is negatively correlated with Silicate with a value of -0.64 . There is a strong positive correlation between Appendicularia and Mero-planktonic larvae with a value of 0.94, as well as with Chaetognatha with a value of 0.85. Salinity has a moderate positive correlation with pH with a value of 0.55, and a weaker positive correlation with Ammonia with a value of 0.50 as shown in Fig. 6. These correlations indicate that nutrient enrichment, particularly ammonia and nitrite accumulation, is closely linked to organic pollution and eutrophic conditions (El-Amier et al. 2016; Hegab et al. 2024). The strong association between Appendicularia, Chaetognatha, and meroplanktonic larvae reflects favourable spawning and feeding conditions in nutrient-rich environments, consistent with previous findings on zooplankton responses to eutrophication gradients (Ger et al. 2014; Pratiwi et al. 2016).

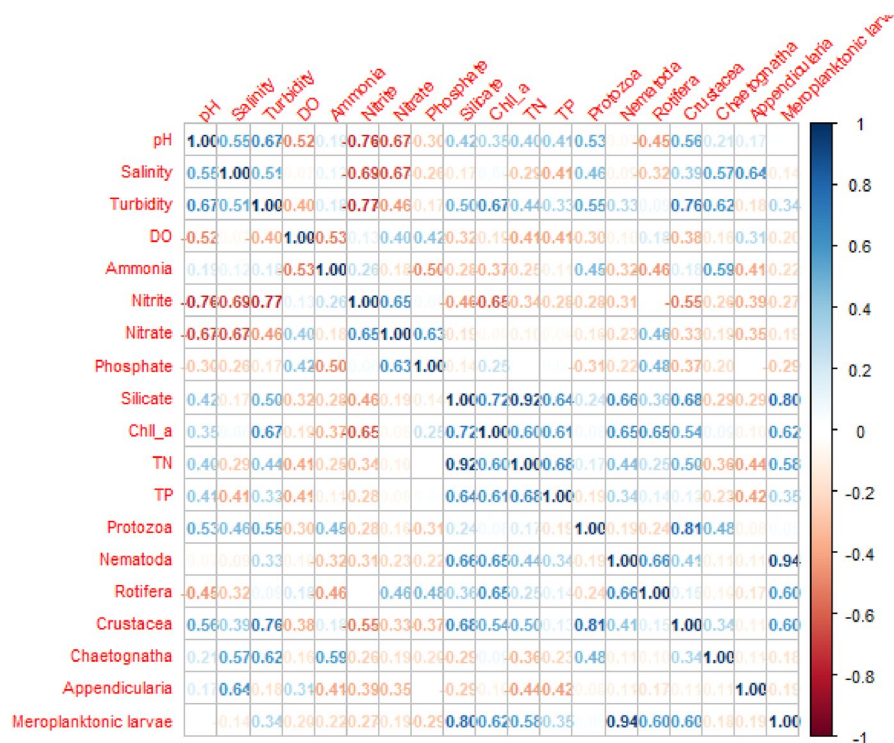


Fig. 6 Correlation among water parameters and zooplankton community

Simulation and prediction of zooplankton community based on water quality

Protozoa play a significant role in ecosystem impacted by eutrophication. In the present study all zooplankton community are affected by some parameters as shown in Fig. 7. Protozoa are affected by pH and turbidity with value of R^2 0.4. All water quality was selected as an indicator of eutrophication status along the study area. Based on correlation analysis, there a positive correlation among chlorophyll and silicate with Nematoda as shown in Table 5. Chlorophyll-a and silicate were positively correlated with nematodes, reflecting nutrient-driven productivity and habitat complexity (Mostafa et al. 2023). Such relationships highlight how eutrophication enhances microbial and benthic interactions within aquatic food webs (Ger et al. 2014).

Factor analysis

The results of factor analysis clearly showed the direct effect of anthropogenic effects in the study area. The first factor that explained 41% of the variance could be named the eutrophication factor. The second factor clearly explains the effect of land-based effluents, which directly affected the values of salinity, pH and nutrients; factor four significant variables that explained less than 10% of variance were also affected by land-based effluents. Factor three,

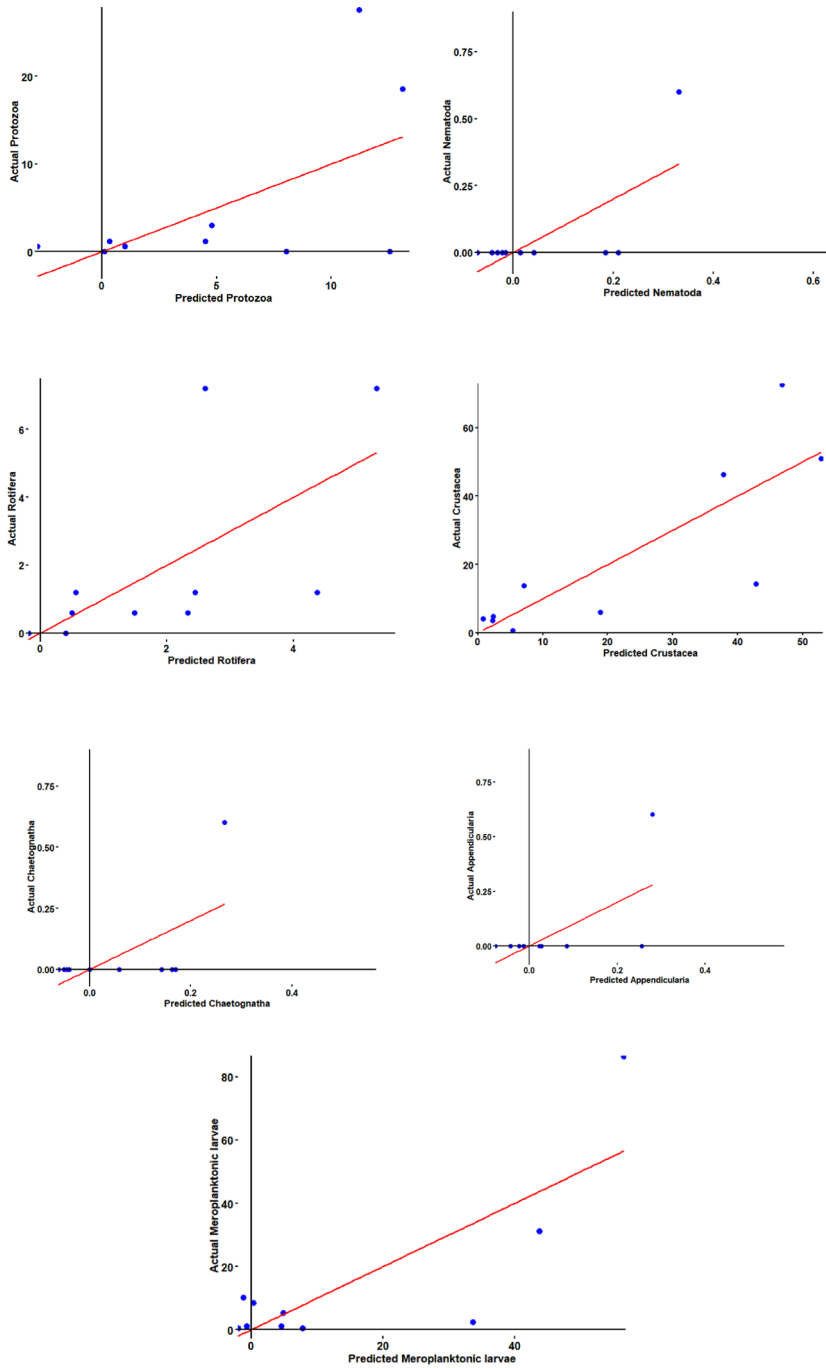


Fig. 7 Correlation among water parameters and zooplankton community

Table 5 Simulation of zooplankton community and water quality

Equation	R^2
$Protozoa = -51.99 + 6.59 * pH + 0.21 * Turbidity$	0.4
$Nematoda = -0.16 + 0 * Chll_a + 0.13 * Silicate$	0.5
$Rotifera = -0.37 + 0.02 * Chll_a$	0.4
$Crustacea = -31.84 + 0.8 * Turbidity + 2.36 * pH + 16.83 * Silicate$	0.6
$Chaetognatha = -0.09 + 0.01 * Turbidity$	0.3
$Appendicularia = -0.08 + 0.01 * Salinity$	0.4
$Meroplanktonic\ larvae = -23.3 + 32.99 * Silicate + 0.03 * Chll_a$	0.6

Table 6 Factor loadings variables on varian rotation with Kaiser normalization

	Rotated component matrix			
	Component			
	1	2	3	4
pH	0.061	0.850	0.364	0.202
Salinity	0.044	0.927	-0.224	0.273
Turbidity	0.542	0.705	0.218	0.024
Dissolved oxygen	-0.143	-0.247	-0.438	-0.679
Ammonia	-0.260	-0.020	0.117	0.870
Nitrite	-0.243	-0.880	-0.141	0.310
Nitrate	0.005	-0.888	0.248	-0.060
Phosphate	-0.168	-0.054	0.132	-0.810
Silicate	0.672	0.186	0.634	-0.049
Chlorophyll_a	0.978	-0.020	0.108	0.114
Chlorophyll_a	0.914	0.265	0.259	-0.100
Chlorophyll_c	0.940	0.122	0.200	0.020
TN	0.512	0.018	0.776	-0.033
TP	0.173	-0.110	0.906	0.161

Table 7 Eigenvalues of the four components

Component	Initial Eigenvalues		
	Total	% of Variance	Cumulative %
1	5.799	41.420	41.420
2	3.165	22.606	64.026
3	2.204	15.740	79.765
4	1.186	8.474	88.239

with a high loading of silicate, total nitrogen, and total phosphorus, can be safely classified as a nutritional factor. (Tables 6 and 7).

Conclusion

This study provides a comprehensive evaluation of eutrophication across five contrasting aquatic habitats in Damietta Governorate, Egypt using an integrated approach that combines in-situ water quality measurements, zooplankton community analyses, and remote

sensing observations. Marked spatial variability in trophic conditions was observed, with Lake Manzala exhibiting the highest eutrophication levels that ranging from eutrophic to hypereutrophic which largely driven by continuous agricultural and industrial inputs. In contrast, the Nile River and the Mediterranean Sea displayed lower trophic states, ranging from mesotrophic to oligotrophic. Zooplankton composition further reflected these environmental gradients. Crustaceans dominated the Mediterranean Sea, emphasizing their central role in marine food-web dynamics. Lake Manzala supported the highest zooplankton density (29520org/m³), characterized by a strong presence of meroplanktonic larvae and crustaceans, consistent with its nutrient-enriched and highly productive conditions. The Barge Canal, where meroplanktons were dominant, appeared to function as an important spawning and recruitment area for benthic organisms. The Nile River was dominated by rotifers, indicating freshwater eutrophication and anthropogenic stress. The Trophic State Index (TSI) model developed using Landsat OLI data demonstrated robust predictive performance ($R^2 = 0.78$), underscoring the potential of satellite-based monitoring for detecting eutrophication hotspots at large spatial scales. Future improvements could be achieved through seasonal sampling and the incorporation of higher-resolution satellite sensors to refine the spatial accuracy of trophic state predictions.

Author contributions Hazem T. Abd El-Hamid carried out the field sampling, laboratory experiments, and sample preparations; conducted the remote sensing analysis; contributed to data analysis, interpretation, and figure preparation; and participated in writing and revising both the first and final versions of the manuscript. Wael S. El-Tohamy participated in field sampling, contributed to figure preparation and data interpretation, and assisted in writing and revising the manuscript. Ahmed S. Shehata contributed to writing; revising and submission of the manuscript to the journal. Hoda F. Zahran contributed to writing and revising the manuscript. Ahmad M. Alzeny carried out the field sampling, data analysis, and interpretation; contributed to conceptualization and experimental design; and participated in writing and revising both the first and final versions of the manuscript.

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Data availability All data generated or analyzed during this study are included in this manuscript and its information files and supplementary file.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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